Domain-Invariant Mitotic Figure Detection for Histology Images

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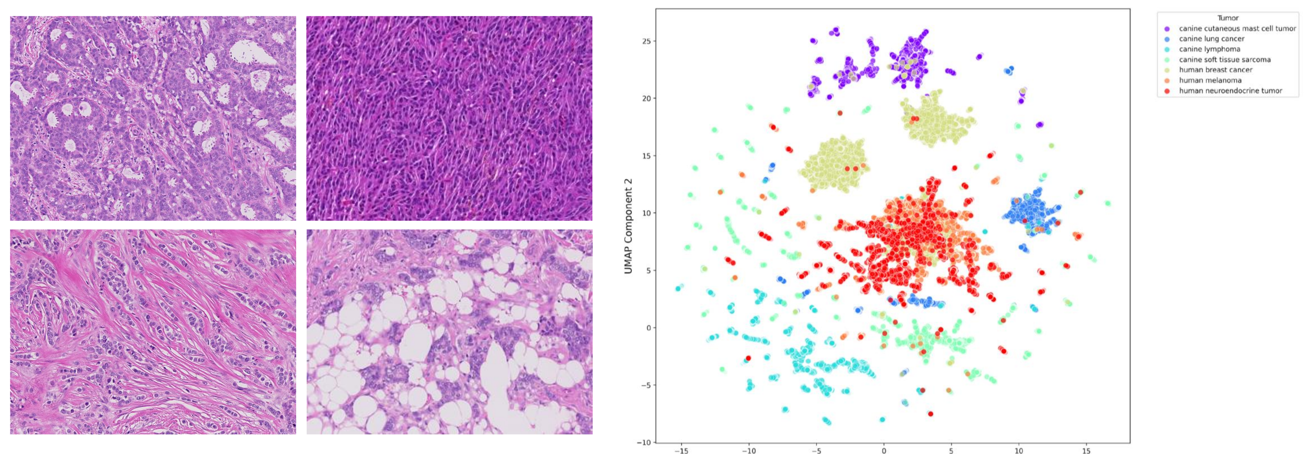
Introduction

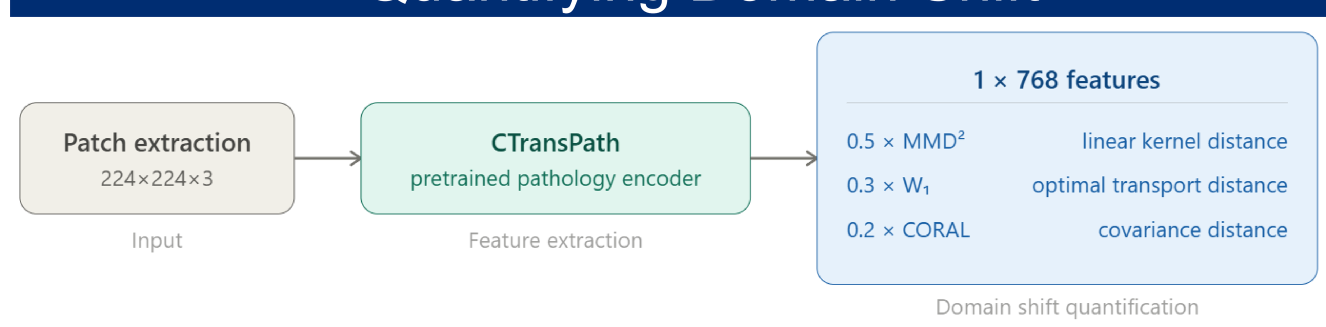


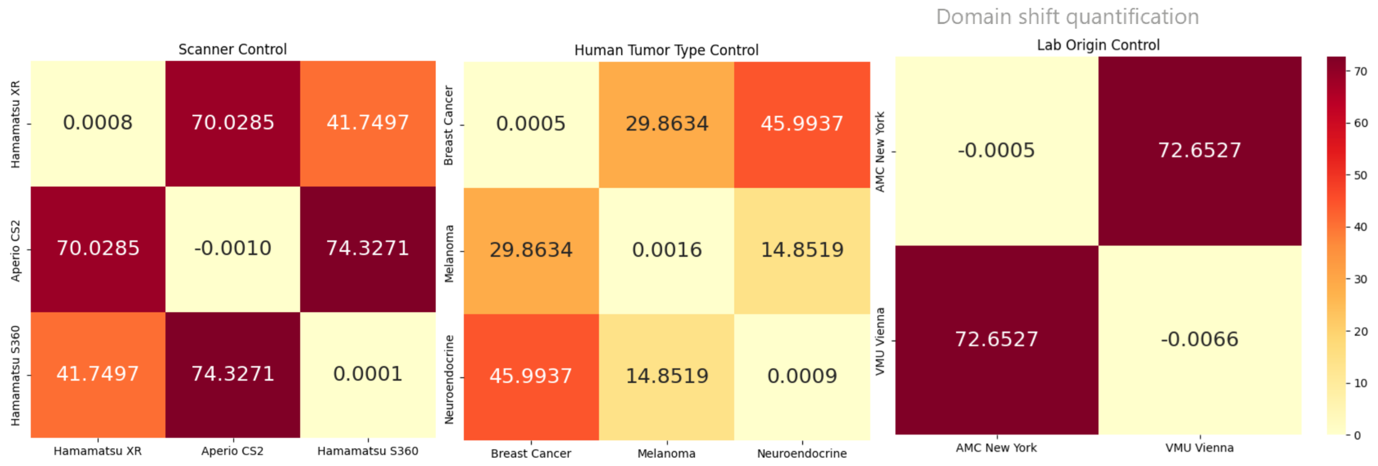
Cancer remains a leading cause of mortality worldwide, accounting for ~10 million deaths in 2020 and projected to reach 16.4 million annually by 2040. Most of the burden falls disproportionately on developing countries due to late-stage detection and costly treatment. Mitotic figure count is a critical indicator of tumor aggressiveness and guides cancer grading in clinical pathology. Digital pathology and automated analysis offer a promising route to reducing dependence on scarce, highly skilled pathologists, making accurate diagnosis more accessible. However, deep learning models for mitotic figure detection suffers from significant performance drops when deployed across different scanners or staining protocols, a challenge known as domain shift.

Goal

To develop a domain-aware deep learning model for mitotic figure detection that generalizes robustly across domains (scanner, tumor type, lab origin, species), supporting automated and accessible cancer grading in clinical pathology.

Quantifying Domain Shift





70–74 Scanner-Induced Variance: Hardware differences (Aperio vs. Hamamatsu) represent a massive source of domain shift, significantly exceeding biological variability.

72.6 Site-Specific Bias: Lab origin (AMC New York vs. VMU Vienna) exhibits an equally extreme divergence, confirming that local staining protocols distort features as much as hardware.

15–46 Biological Consistency: In contrast, distances between Human Tumor Types are significantly lower, proving non-biological noise is the dominant factor in baseline shift.

Validation

This high baseline divergence validates the need for a domain-invariant framework to "scrub" these non-diagnostic signatures.

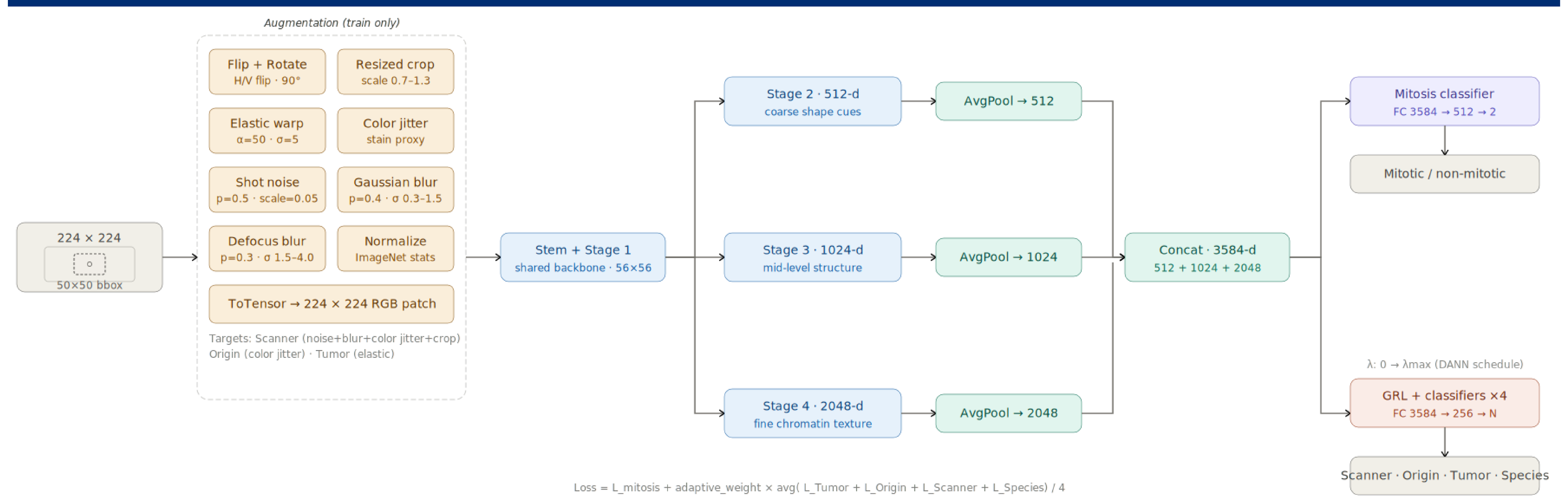
Distance Metrics

$$MMD^2 = \frac{1}{n(n-1)}\sum_i \sum_{j\neq i} x_i^T x_j + \frac{1}{m(m-1)}\sum_i \sum_{j\neq i} y_i^T y_j - \frac{2}{nm}\sum_i \sum_j x_i^T y_j$$ Kernel Distance

$$L_{CORAL} = \frac{1}{4d^2}\|C_{source}(x) - C_{target}(x)\|_F^2$$ Covariance Distance

$$W_1(source, target) = \int_{-\infty}^{+\infty} |CDF_{source}(x) - CDF_{target}(x)|\,dx$$ Optimal Transport

Model Architecture



Features are extracted from a ResNet50 backbone at multiple depths (Layers 2, 3, and 4) and concatenated into a 3584-d vector to capture both local chromatin texture and global spindle geometry. A Gradient Reversal Layer (GRL) facilitates simultaneous adversarial training across four domain heads (Scanner, Lab Origin, Species, and Tumor Type) to ensure feature invariance.

Gradient Reversal Layer (GRL)

To achieve feature invariance, we utilize a Gradient Reversal Layer (GRL) that treats domain classification as an adversarial task, forcing the backbone to 'forget' scanner and species-specific signatures. It is defined by a pseudo-function that behaves differently during the two phases of backpropagation:

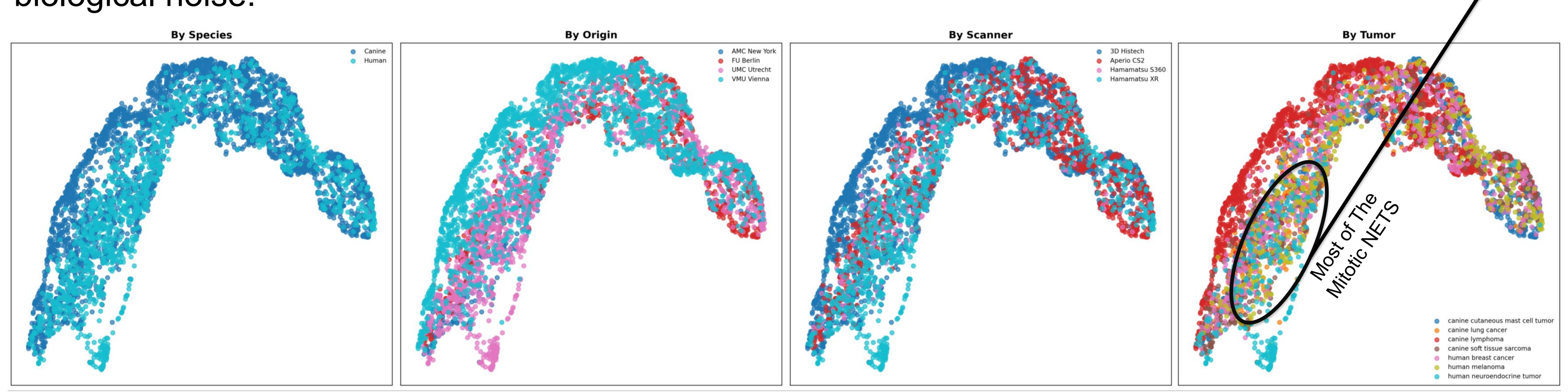
- **Forward Propagation:** Acts as an identity mapping to allow the flow of features for classification. $f(x) = x$
- **Backward Propagation:** Flips the gradient and scales it by a factor of λ, forcing the backbone to learn representations that are indistinguishable to the domain classifiers. $\nabla\mathcal{L} = -\lambda\frac{\partial\mathcal{L}}{\partial x}$

To prevent the adversarial signal from interfering with early-stage feature extraction, a sigmoid-based schedule for the adaptation parameter λ was implemented. This allows the model to prioritize mitosis classification accuracy before introducing the domain-invariance constraint:

$$\lambda_p = \frac{4}{1+\exp(-\gamma \cdot p)} - 1$$

- **γ (Hyperparameter):** Set to 10 to control the steepness of the ramp.
- **p (Training Progress):** Defined as current epoch / total epochs ($0 \to 1$).

UMAP Embeddings of the 3584-d Feature Vector shows homogeneous mixing across all domain aspects showing that the GRL successfully disentangled diagnostic features from sensor-level and biological noise.

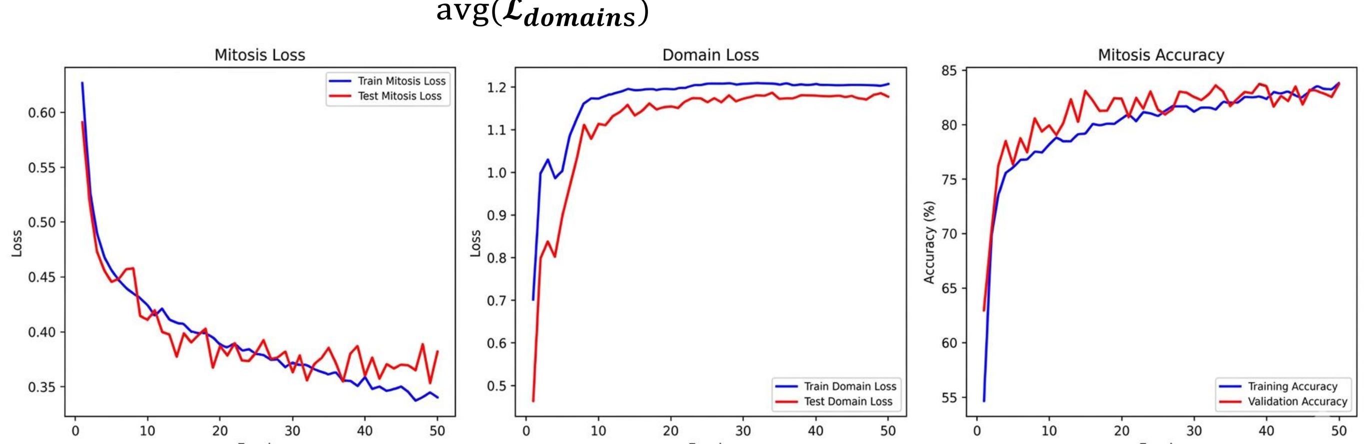


Combined Loss Function

We optimize the following multi-objective loss:

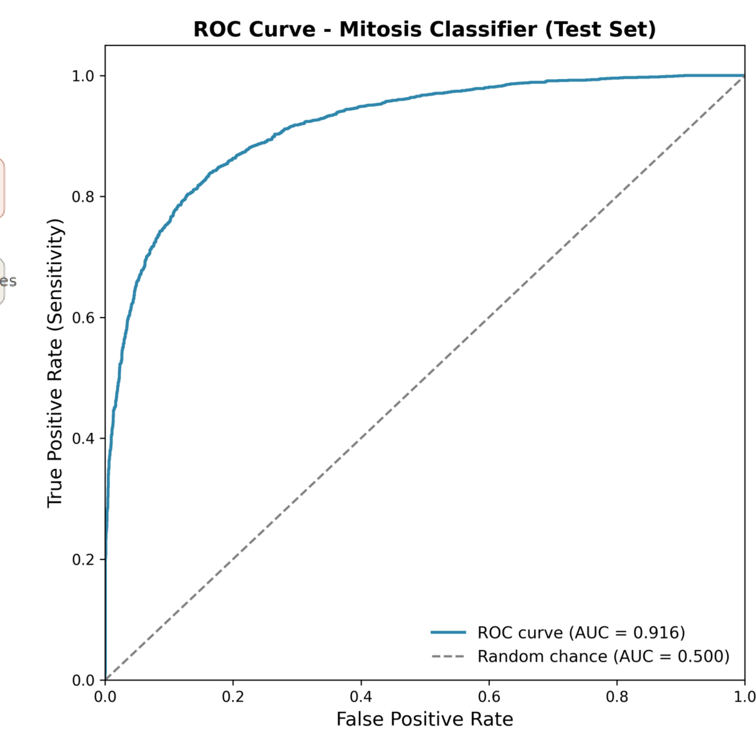
$$\mathcal{L}_{total} = \mathcal{L}_{mitosis} + \alpha\frac{1}{4}\sum_{i=1}^{4}\mathcal{L}_{domain_i}$$

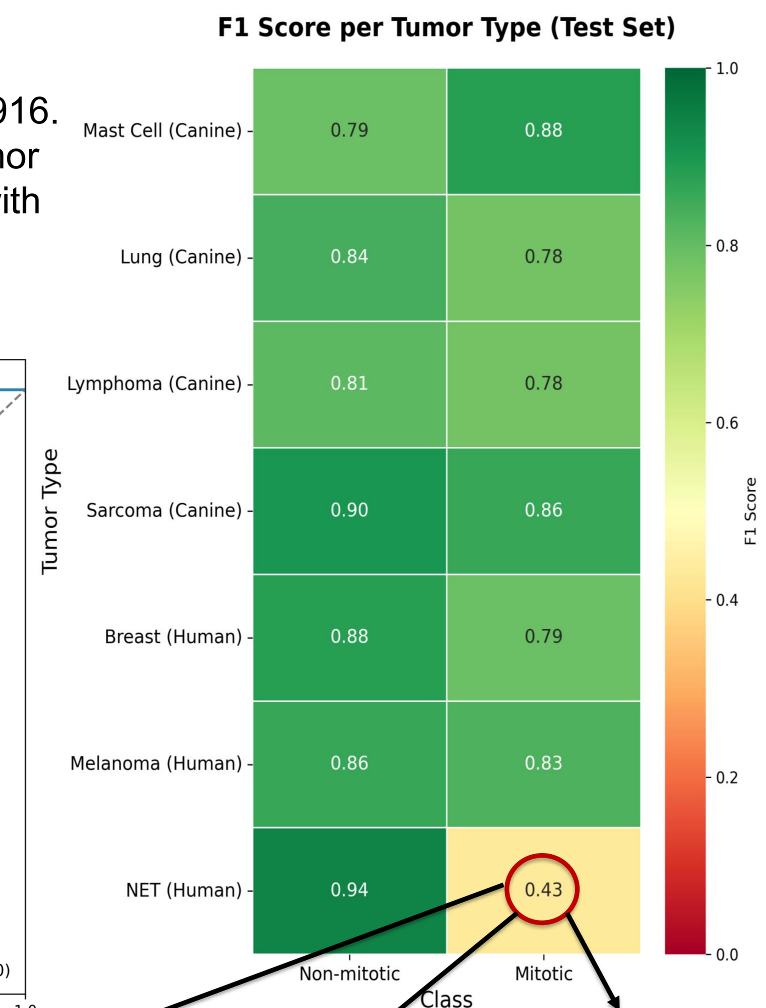
- **$\mathcal{L}_{mitosis}$**: Weighted Cross-Entropy (Mitotic weight: 2.0) to handle class imbalance.
- **AdaptiveWeight:** A momentum-based running average that keeps the scale of domain gradients proportional to the mitosis task: $\alpha \approx \frac{\mathcal{L}_{mitosis}}{\mathrm{avg}(\mathcal{L}_{domains})}$

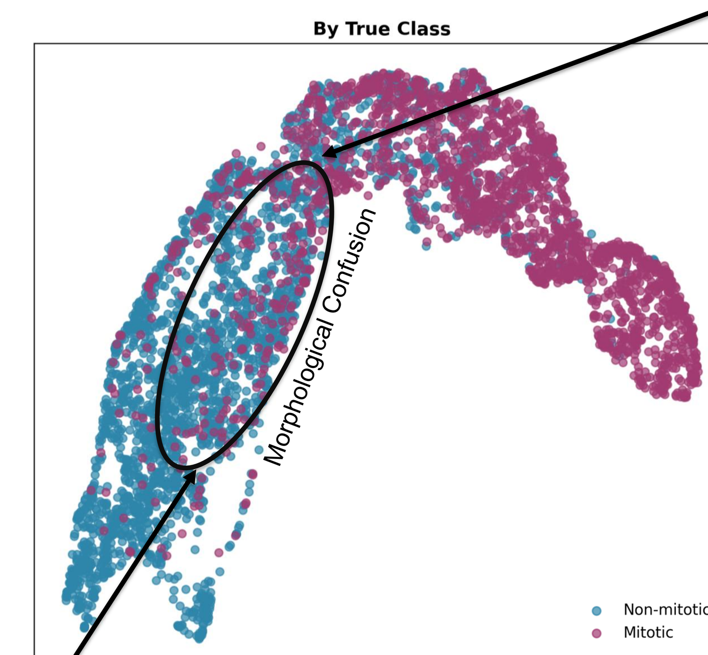


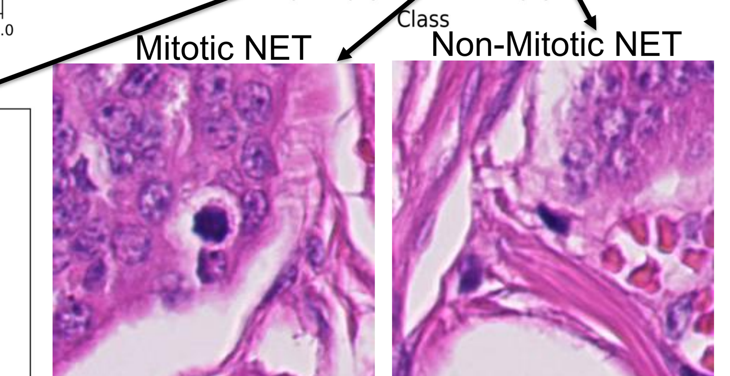
Results

Global Diagnostic Performance: The ROC curve shows high diagnostic discriminative power with an AUC of 0.916. (Right) F1-score breakdown across tumor types indicates robust generalization, with major performance drops in specific neuroendocrine tumor (NET) subsets.









Failure Case Analysis: (Left) Mapping the lowest-performing class (Mitotic NET) back to the latent space reveals significant overlap with non-mitotic clusters. (Right) Visual inspection suggests that hyperchromatic, dense nuclei in NETs act as high-confidence false positives, indicating a limit to morphological invariance.

Conclusion

The model achieves a high 0.916 AUC, validated by the clear separation of mitotic features in the latent space topology. While the multi-scale adversarial framework successfully eliminates hardware and site-specific noise, analysis of the NET cohort reveals a specific performance ceiling. The morphological confusion observed on the manifold suggests that our use of Elastic Transforms and Blur/Noise augmentations, while essential for domain robustness, may inadvertently scrub the fine-grained chromatin textures needed to distinguish biological mimics. This indicates that the remaining bottleneck is a trade-off between generalization and the textural sensitivity required to resolve extreme morphological ambiguity.

Takeaway

Multi-scale adversarial training achieves 0.916 AUC by eliminating scanner-specific noise as a performance ceiling. This shifts the remaining budget entirely onto genuine morphological ambiguity: hyperchromatic NET nuclei that challenge even expert pathologists.

References

Aubreville, M., Wilm, F., Stathonikos, N. et al. A comprehensive multi-domain dataset for mitotic figure detection. Sci Data 10, 484 (2023). https://doi.org/10.1038/s41597-023-02327-4

Foi, A., Trimeche, M., Katkovnik, V., & Egiazarian, K., "Practical Poissonian-Gaussian Noise Modeling and Fitting for Single-Image Raw-Data," in IEEE Transactions on Image Processing, vol. 17, no. 10, pp. 1737-1754, Oct. 2008, doi: 10.1109/TIP.2008.2001399.

Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., Marchand, M., & Lempitsky, V.S. (2015). Domain-Adversarial Training of Neural Networks. Journal of machine learning research.

Shabbir, B., Saleem, S., Aleem, I., Babar, N., Farooq, H., Loya, A., & Naveed, H. (2024). Low-Cost Histopathological Mitosis Detection for Microscope-acquired Images. AMIA Joint Summits on Translational Science proceedings. AMIA Joint Summits on Translational Science, 2024, 409–418.t